

Vectors



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But the idea is the same!

Definition

Vector: A vector is an entity with **direction** and **length** (or magnitude!)

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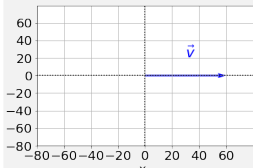
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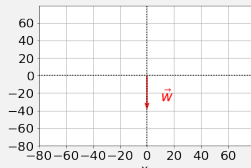
Example

A car traveling 60 miles per hour directly East.



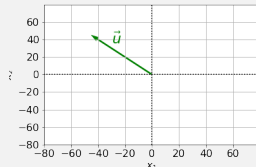
$$\mathbf{v} = \vec{v} = \begin{bmatrix} 60 \\ 0 \end{bmatrix}$$

A car traveling 40 miles per hour directly South.



$$\mathbf{w} = \vec{w} = \begin{bmatrix} 0 \\ -40 \end{bmatrix}$$

A car traveling 64 miles per hour directly North West.



$$\mathbf{u} = \vec{u} = \begin{bmatrix} -\frac{64}{\sqrt{2}} \\ \frac{64}{\sqrt{2}} \end{bmatrix}$$

Vectors in $\mathbb{R}^n$

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If the vector $\mathbf{v}$ has n different real numbers, then we say that $\mathbf{v} \in \mathbb{R}^n$, and

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

Vector Equality

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- ▶ They have the **same direction and length** or
- ▶ All entries of the two vectors are the same:

$$v_k = w_k \text{ for all } 1 \leq k \leq n$$

Vector Arithmetic (Addition)

Definition

We define adding two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ as $\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{bmatrix}$. Note: This means that we can only add vectors with the same dimensions together! And the result will have the same dimension!

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Adding the following vectors

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$\mathbf{u} = \begin{bmatrix} 0 \\ -1 \\ 5 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 7 \\ -5 \\ 10 \end{bmatrix}$$

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$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} 1 + 4 \\ 2 + 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} 0 + 7 \\ -1 + -5 \\ 5 + 10 \end{bmatrix} = \begin{bmatrix} 7 \\ -6 \\ 15 \end{bmatrix}$$

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Computing the scalar product of the following

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Example

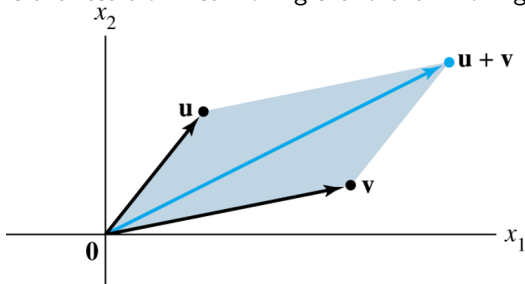
Computing the scalar product of the following

$$c = -3, \mathbf{v} = \begin{bmatrix} 2 \\ -5 \end{bmatrix}, \quad c\mathbf{v} = \begin{bmatrix} -3 \cdot 2 \\ -3 \cdot -5 \end{bmatrix} = \begin{bmatrix} -6 \\ 15 \end{bmatrix}$$

Geometric Descriptions of Vectors in $\mathbb{R}^2$

We can think of the vector $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in \mathbb{R}^2$ as a directed line segment (an arrow).

Consider two vectors $\mathbf{u}$ and $\mathbf{v} \in \mathbb{R}^2$. The sum $\mathbf{u} + \mathbf{v}$ is the result of first moving $\mathbf{u}$ and then moving $\mathbf{v}$.



Consider the vector $\mathbf{u} \in \mathbb{R}^2$ and nonzero scalar c . The scalar product $c\mathbf{u}$ gives a vector parallel to $\mathbf{u}$ whose magnitude has been scaled by a factor of c .

- ▶ If $c > 0$, then $c\mathbf{u}$ has the same direction as $\mathbf{u}$.
- ▶ If $c < 0$, then $c\mathbf{u}$ has the opposite direction as $\mathbf{u}$.
- ▶ If $|c| < 1$, then $c\mathbf{u}$ is a compression of $\mathbf{u}$.
- ▶ If $|c| > 1$, then $c\mathbf{u}$ is a stretching of $\mathbf{u}$.

Parallelogram Rule for Addition: If $\mathbf{u}$ and $\mathbf{v}$ are in $\mathbb{R}^2$, then the sum $\mathbf{u} + \mathbf{v}$ is the diagonal of the parallelogram spanned by $\mathbf{u}$ and $\mathbf{v}$.

Properties of Vector Arithmetic

The **zero vector** is defined as the vector in $\mathbb{R}^n$ with no magnitude, and it is denoted by $\mathbf{0}$ or $\vec{0}$.

As a result of this definition, it follows that $\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$.

For all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in $\mathbb{R}^n$ and all scalars c and d :

- | | |
|--|--|
| (i) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ | (v) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$ |
| (ii) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ | (vi) $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$ |
| (iii) $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$ | (vii) $c(d\mathbf{u}) = (cd)\mathbf{u}$ |
| (iv) $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ | (viii) $1\mathbf{u} = \mathbf{u}$ |

Linear Combinations of Vectors

Definition

Linear Combination: A linear combination of vectors $\mathbf{v}_1, \dots, \mathbf{v}_p \in \mathbb{R}^n$ with weights $c_1, \dots, c_p \in \mathbb{R}$ is the vector $\mathbf{y}$ given by

$$\mathbf{y} = c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \sum_{k=1}^p c_k\mathbf{v}_k$$

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$$2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + -3 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

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Linear Combinations of Vectors

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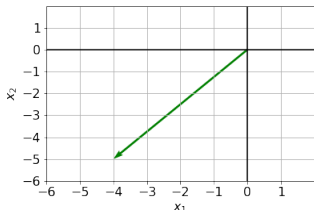
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Giving Directions in a Grid

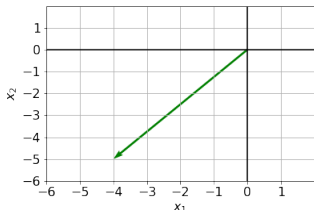


Example

Find constants x_1 and x_2 such that

$$\begin{bmatrix} -4 \\ -5 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

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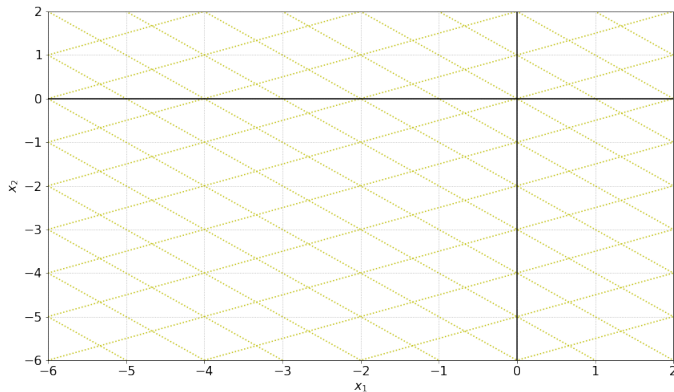
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The vectors $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are the **standard column vectors**.

Giving Directions in a Different System



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Find constants x_1 and x_2 such that

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Solving for the Weights

First, let's simplify our statement!

$$\begin{bmatrix} -4 \\ -5 \end{bmatrix} = x_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2x_1 - x_2 \\ x_1 + x_2 \end{bmatrix}$$

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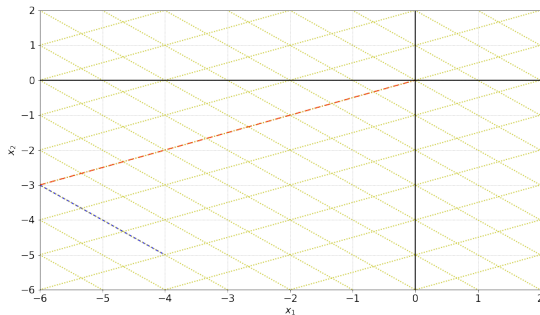
$$-5 = 1x_1 + 1x_2$$

Gaussian Elimination can solve this!

$$\left[\begin{array}{cc|c} 2 & -1 & -4 \\ 1 & 1 & -5 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & -3 \\ 0 & 1 & -2 \end{array} \right]$$

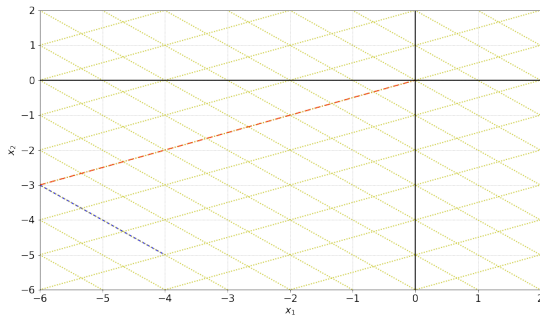
So, $(x_1, x_2) = (-3, -2)$. Or we can say $\mathbf{x} = \begin{bmatrix} -3 \\ -2 \end{bmatrix}$

What Our Directions Look Like



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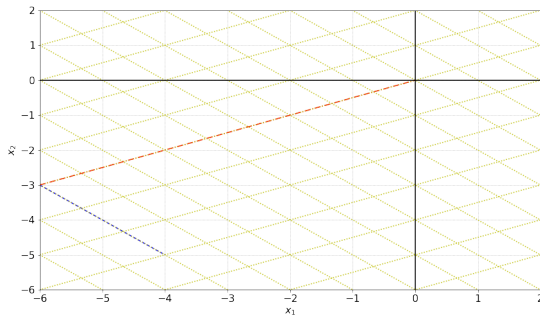
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$$\mathbf{x} = \begin{bmatrix} -3 \\ -2 \end{bmatrix}$$

What about an arbitrary vector $\mathbf{w} \in \mathbb{R}^2$? Can we find an $\mathbf{x}$ such that

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = x_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Span of a Set of Vectors

Definition

Span: Let $\mathbf{v}_1, \dots, \mathbf{v}_p \in \mathbb{R}^n$. Then we define $\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_p)$ to be the subset of $\mathbb{R}^n$ containing all linear combinations of $\mathbf{v}_1, \dots, \mathbf{v}_p$.

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$$\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_p) = \left\{ \mathbf{y} \in \mathbb{R}^n \mid \mathbf{y} = \sum_{k=1}^p c_k \mathbf{v}_k \text{ for some } c_1, \dots, c_p \in \mathbb{R} \right\}$$

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How do we test if $\mathbf{y} \in \text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_p)$?

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How do we test if $\mathbf{y} \in \text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_p)$?

Solution

Solve the system

$$\left[\begin{array}{ccc|c} \mathbf{v}_1 & \cdots & \mathbf{v}_p & \mathbf{y} \end{array} \right]$$

Span Example

Example

Describe the span of the vectors $\mathbf{v}_1, \mathbf{v}_2$ given by

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$$

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Solution

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 = c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} c_1 \\ 2c_2 \\ c_1 + c_2 \end{bmatrix}$$

So,

$$\text{Span}(\mathbf{v}_1, \mathbf{v}_2) = \left\{ \mathbf{y} \in \mathbb{R}^3 \mid \mathbf{y} = \begin{bmatrix} c_1 \\ 2c_2 \\ c_1 + c_2 \end{bmatrix} \text{ for some } c_1, c_2 \in \mathbb{R} \right\}$$

Span of a Set of Vectors Practice

Describe the span of the following vectors

1. $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

2. $\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -4 \\ -2 \end{bmatrix}$

3. $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

4. $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$

Span of a Set of Vectors Practice Solutions

Describe the span of the following vectors

1. $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$

$$\text{Span}(\mathbf{v}_1) = \left\{ \mathbf{y} \in \mathbb{R}^3 \mid \mathbf{y} = \begin{bmatrix} 0 \\ c_1 \\ c_1 \end{bmatrix} \text{ for some } c_1 \in \mathbb{R} \right\}$$

2. $\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -4 \\ -2 \end{bmatrix}$

$$\text{Span}(\mathbf{v}_1, \mathbf{v}_2) = \left\{ \mathbf{y} \in \mathbb{R}^2 \mid \mathbf{y} = \begin{bmatrix} 2c_1 - 4c_2 \\ c_1 - 2c_2 \end{bmatrix} \text{ for some } c_1, c_2 \in \mathbb{R} \right\}$$

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$$\text{Span}(\mathbf{v}_1, \mathbf{v}_2) = \left\{ \mathbf{y} \in \mathbb{R}^3 \mid \mathbf{y} = \begin{bmatrix} c_1 \\ c_2 \\ -2c_1 - c_2 \end{bmatrix} \text{ for some } c_1, c_2 \in \mathbb{R} \right\}$$

More Span Practice

Determine if $y = \begin{bmatrix} 4 \\ 5 \\ -1 \end{bmatrix}$ is in $\text{Span} \left(\begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right)$

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Perform Gaussian Elimination on the augmented matrix

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$$\left[\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & 5 \\ -2 & -1 & -1 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & 5 \\ 0 & 0 & 12 \end{array} \right]$$

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$$\left[\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & 5 \\ -2 & -1 & -1 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & 5 \\ 0 & 0 & 12 \end{array} \right]$$

No!

Systems with Infinite Solutions

Recall for the system,

$$1x_1 + 2x_2 + x_3 = -2$$

$$1x_1 + 3x_2 - 2x_3 = 1$$

Our answer is

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$$\begin{cases} x_1 &= -8 - 7x_3 \\ x_2 &= 3 + 3x_3 \\ x_3 &\text{is free} \end{cases}$$

Systems with Infinite Solutions

Recall for the system,

$$1x_1 + 2x_2 + x_3 = -2$$

$$1x_1 + 3x_2 - 2x_3 = 1$$

Our answer is

$$\begin{cases} x_1 &= -8 - 7x_3 \\ x_2 &= 3 + 3x_3 \\ x_3 &\text{is free} \end{cases}$$

But writing this can be cumbersome!

Using Vectors!

We recall from earlier that we can write out our answer in terms of vectors!

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Span of a Set of Vectors Conceptual Practice

Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{w}$ be vectors in $\mathbb{R}^n$. Prove that if $\mathbf{w} \in \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$, then $\text{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{w}) = \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$